

Behavior Engineering Using Quantitative Reinforcement Learning Models

Corresponding Author: Dr Ohad Dan

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The paper presents the results of one of the two choice engineering competitions proposed by Dan and Loewenstein (2019). I was one of the reviewers of the 2019 paper, and thought that the choice engineering competition idea is a creative method that can advance the cognitive, behavioral and social sciences. The results presented in the current paper support my initial optimistic belief. In particular, the finding the popular QL models were outperformed by a model that assume a very different process is very important. As the authors explain (here and in their previous research) QL models have many attractive features, but if they cannot perform well in the current simple engineering task, it is possible that they are less useful than it is commonly assumed.

I think that the paper can be improved by weakening some of the assertions concerning the difference between choice engineering and choice architecture. Here are some suggestions:

1. I propose that the author will note (a footnote might suffice) that their definition of choice architecture differs from the common definitions. The explicit definition by Thaler and Sunstein (2008) involves manipulations that do not change the incentive structure in a significant way. The common applications involve the use of experiments to check which of several “behavioral insights” is more effective.

2. I think that the authors should pay more attention to the second competition they run. The finding that no one of the competitors used a model is not a problem: it is an important finding. It suggests that extant learning research does not pay enough attention to learning in interactive settings. I suggest that the author will include a section (maybe in the supplementary material) that describe this competition, and invite the winner of that competition to be a coauthor of the current paper.

3. I am not sure that I understand the post hoc iterative process analysis that leads 69% success rate. From what I do understand, this method sounds similar to the way practitioners use choice architecture (as I noted above): They start with some behavioral insight (primacy and distinct diminishing exploration functions), and then run experiments to compare the distinct diminishing exploration functions.

Please clarify, maybe by presenting the sequence, and the logic behind the changes between iterations. If I am correct, and this method is more architecture-like, this fact should be explained. When it is possible to experimentally test many conditions: architecture-like methods can help. Of course, it is possible that starting the iteration with CATIE would lead to better results.

4. Please clarify the difference between the difference between the Average choice probability ($E[p]$), and the Log-likelihood ($E[\log(p)]$) analysis in Figure 4. Can this difference be explained by the fact that the prediction of CATIE are too extreme?

5. Please discuss the relationship between choice engineering competitions, and choice prediction competition. It is easy to see that the engineering approach you used are less efficient (they require running new experiments to each model). This point should be noted, and it will also be good to clarify the advantages of the engineering approach.

Reviewer #2

(Remarks to the Author)

This paper reports the results of a unique and creative "choice engineering" competition, which evaluates models based on how well they can induce human subjects to follow particular choice patterns. I previously reviewed the call for entries. I think the results are interesting and definitely worth reporting, provided the authors can address the points below.

Major comments:

There is not enough information about the models. Minimally, the models should be described at a level of detail which would allow them to be reproduced. Ideally, software for the models themselves and the procedures for generating reward sequences would be available on the competition website.

There is a lot of emphasis on CATIE (the winning model), but I'm not sure how justified this is given that the 2nd place model produced indistinguishable performance.

I'm surprised that the model comparison results in Fig. 4 differ so much between $E[p]$ and $E[\log(p)]$, given that $\log(p)$ is a monotonic function of p . Can the authors explain this?

The number of participants used to evaluate each model differs quite a bit. For example, Model 11 had 87 subjects, whereas Model 0 had 549 subjects. I'm sorry if I missed it, but I didn't see any justification of these differences. I suggest that the authors provide some evidence that these sample sizes are sufficient for reliable model comparison.

I was asked to comment on the dynamic track data, in particular whether it's reported in enough detail and whether it's worth incorporating into the manuscript. In my opinion, I think it makes sense to leave it where it is in the supplement. With respect to transparency, the reporting suffers the same issue as for the static track, namely the models are not described in sufficient detail to be reproducible.

Minor comments:

Line 140: "subject's" -> "the subject's"

Fig 4: I assume the y-axis of panel b is probability x 100?

Reviewer #3

(Remarks to the Author)

Thank you for sending me this paper for review. In it, Dan, Plonsky and Loewenstein describe the results of the Choice Engineering Competition (announced in Nat Comms in 2019) designed to show that quantitative models can be used to shape human behaviour. Briefly, the asked contestants to place a limited number of rewards such as to maximally bias behaviour. They obtained 11 submissions. They then tested each of these 11 submitted reward schedules on real human participants and examined them using two different types of prominent choice / learning models: a Q-learning model and a CATIE model. They distinguish choice engineering from choice architecture, with the former relying on an explicit model of choice; and the key findings are that contestants were able to design effective reward schedules; that CATIE outperformed Q learning in terms of explaining the human data (at least on some measures); and that CATIE also outperformed Q learning as a basis for designing effective reinforcement schedules.

There is much to like about the paper. It is an interesting and timely question which speaks to an important aspect of generalizability of a large swathe of in-lab studies. The project was clearly announced and the results should be of broad interest.

Overall, the paper is well written, but it would be good to be somewhat clearer about the nomenclature at several points, e.g. the distinction between architect and engineer; the distinction between the various datasets (lab, online, previous, new...); it would be helpful to have an overview figure over the process / datasets; and it would be good to have a brief table in the main paper describing the submissions as otherwise most of the figures are difficult to read.

Minor:

- Figure 3 seems to refer to 12 datasets (0-11), but there are only 11 elsewhere.

- some submissions refer explicitly to examining the results on prolific, which is also what the authors did themselves. it seems that this deserves some more attention - why did this not work?

- line 239: CATIE is stated to achieve 72.4 $\pm$ 0.1, which is different from line 165 where it is 72.7 $\pm$ 0.1.

- figure 4: legend refers to light red, but there is no such colour in the figure. Please clarify error bars - quantiles would be more informative and allow relationship between b and c to be clearer.

- supplement table with models: hetro -> hetero,

- supplement fig s2: scehulde -> schedule

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The revised draft addresses the shortcoming of the original submission, and nicely summarizes an important project. I recommend acceptance for publication.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

I am satisfied with the revision. I was also asked to comment on the inclusion of Schedule 0. I thought the authors gave a reasonable justification for its inclusion.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

Thank you very much for the responses and the updated paper. I find this substantially improved and do not have further comments.

(Remarks on code availability)

Open Access This Peer Review File is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.

The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain

permission directly from the copyright holder.

To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

We are grateful to the reviewers and the editorial team for their insights which helped us improve the manuscript. We believe that the revised version provides a clearer and more comprehensive presentation of the competition, its results, and its implications.

In response to the reviewers' and editorial team's comments, we made several major changes to the manuscript including: (a) rewriting of the introduction in a way that better clarifies the difference between the architecture and engineering approaches. (b) A reorganization of the Results chapter. We now begin with a presentation of the competition results, including a more elaborative presentation of the principle used to design the reward schedules. Only then, we follow with additional analyses, such as the evaluation of the performance of the winning model, CATIE. In addition to a more streamlined presentation of our findings, this presentation better reflects the role of Schedule 0 and the way it was constructed. (c) We enriched the Discussion with an additional section describing the comparison of choice engineering competitions with choice prediction competitions. (d) Finally, we revised several of the figures, and in response to the reviewers' comments added two new supplementary figures (Fig. S4, and S5).

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

The paper presents the results of one of the two choice engineering competitions proposed by Dan and Loewenstein (2019). I was one of the reviewers of the 2019 paper, and thought that the choice engineering competition idea is a creative method that can advance the cognitive, behavioral and social sciences. The results presented in the current paper support my initial optimistic belief. In particular, the finding the popular QL models were outperformed by a model that assume a very different process is very important. As the authors explain (here and in their previous research) QL models have many attractive features, but if they cannot perform well in the current simple engineering task, it is possible that they are less useful than it is commonly assumed.

I think that the paper can be improved by weakening some of the assertions concerning the difference between choice engineering and choice architecture. Here are some suggestions:

1. I propose that the author will note (a footnote might suffice) that their definition of choice architecture differs from the common definitions. The explicit definition by Thaler and Sunstein (2008) involves manipulations that do not change the incentive structure in a significant way. The common applications involve the use of experiments to check which of several “behavioral insights” is more effective.

Thank you for the comment, which motivated us to rewrite the entire Introduction section in order to clarify this point. Specifically, we now write:

“Sunstein and Thaler coined the term choice architecture in the framework of ‘nudges’, behavioral interventions that “alters people’s behavior in a predictable way without forbidding any options or significantly changing their economic incentives”³. Sunstein and Thaler’s architecture approach is commonly guided by a qualitative and not quantitative, understanding of behavior, and the effectiveness of the interventions can only be tested empirically. Therefore, we adopt the term choice architecture here to describe the use of qualitative psychological principles in order to influence choices without changing the alternatives’ objective values. This is in contrast to an approach that is based on quantitative models, which we term choice engineering.”

2. I think that the authors should pay more attention to the second competition they run. The finding that no one of the competitors used a model is not a problem: it is an important finding. It suggests that extant learning research does not pay enough attention to learning in interactive settings. I suggest that the author will include a section (maybe in the supplementary material) that describe this competition, and invite the winner of that competition to be a coauthor of the current paper.

In Table S2 of the supplementary section we describe the results of the dynamic competition.

We agree that learning in an interactive setting is an interesting question that has not received enough attention. However, this finding about the submissions is anecdotal. An equally valid interpretation of the observation that no one of the competitors used a model is a failure on our side to properly advertise the competition.

After thorough consideration and discussing this point with the Editors, we acknowledge the winners of the dynamic competition by name in the presentation of the dynamic track but we have decided not to invite them to coauthor the paper.

3. I am not sure that I understand the post hoc iterative process analysis that leads 69% success rate. From what I do understand, this method sounds similar to the way practitioners use choice architecture (as I noted above): They start with some behavioral insight (primacy and distinct diminishing exploration functions), and then run experiments to compare the distinct diminishing exploration functions.

Please clarify, maybe by presenting the sequence, and the logic behind the changes between iterations. If I am correct, and this method is more architecture-like, this fact should be explained. When it is possible to experimentally test many conditions: architecture-like methods can help. Of course, it is possible that starting the iteration with CATIE would lead to better results.

Indeed, the process that generated sequence 0 was similar to the way that practitioners may try to optimize a schedule, if given enough time and resources. It included testing schedules on human participants and using the behavioral results of each iteration to construct a new schedule. Our method was not principled. Rather, it

was based on a combination of data analysis and intuition. Consequently, we are not claiming that the method that we used was efficient or optimal. Only, we used this schedule to show that there exists a schedule for which the bias is 69%. Therefore, we do not see value in presenting the sequence of schedules in the manuscript.

In the manuscript, we now describe the process and the logic behind it in the section “Schedule tuning” of the Results section.

“We cannot find the optimal schedule or calculate the maximally attainable bias, but it is possible to use a data-driven approach in order to improve a reward schedule (constructed by an engineer, an architect or even chosen randomly). Such an approach involves iteratively testing various schedules on human subjects and refining the schedules based on their performance. The ability of AlphaZero to achieve superhuman performance in games such as chess and Go without incorporating any explicit human knowledge about these games is a testament to the strength of this approach^{11,12}. With sufficient resources, this approach can be successful even without using any domain knowledge about the computational principles or algorithms underlying operant learning in humans. We therefore used it to construct a benchmark, relative to which the performance of the competing schedules can be compared.

Specifically, we started by testing subjects on the naive reward schedule of Fig. 1c. We then calculated the empirical per-trial probability of choosing each of the alternatives. With these probabilities, we used our intuition to reallocate some of the rewards, with a goal of maximizing the harvest of rewards from Alternative 1 and minimizing the harvest of rewards from Alternative 2. We repeated this process three times, until the changes that we made to the reward schedule were detrimental to performance (bias in favor of Alternative 1). This process resulted in Schedule 0, that is depicted in Fig. S1 (top). Notably, we executed this iterative process prior to the competition so that we will be able to test this schedule at the time of the competition. We found that across subjects, the elicited bias was 69.0% $\pm$ 0.5%. This result is a lower bound of the maximally attainable bias. Given that a random schedule induces a bias of 50%, the effectiveness of the winning schedule is lower than $\frac{64.3\%-50\%}{69.0\%-50\%} = 75\%$, but we speculate that it is not much lower.”

4. Please clarify the difference between the difference between the Average choice probability ($E[p]$), and the Log-likelihood ($E[\log(p)]$) analysis in Figure 4. Can this difference be explained by the fact that the prediction of CATIE are too extreme?

Thank you for highlighting this point. Indeed, in the CATIE model, there is a larger fraction of actions to which the model assigns a small probability. This can explain

the difference between average choice probability and log-likelihood. We clarify this point by adding a new supplementary figure (new Fig. S4).

The reviewer's comment also motivated us to take a closer look at the predictive power of CATIE and the QL models. In the revised manuscript, we also compare the predictive powers of the models per schedule (new Fig. S5). Intriguingly, we found that CATIE was significantly better in predicting behavior in some schedules, but not in others. We now comment on this observation in the revised manuscript.

5. Please discuss the relationship between choice engineering competitions, and choice prediction competition. It easy to see that the engineering approach you used are less efficient (they require running new experiments to each model). This point should be noted, and it will also be good to clarify the advantages of the engineering approach.

In response to this comment, we added a full section in the Discussion section entitled "Engineering competitions vs. prediction competitions".

Reviewer #2 (Remarks to the Author):

This paper reports the results of a unique and creative "choice engineering" competition, which evaluates models based on how well they can induce human subjects to follow particular choice patterns. I previously reviewed the call for entries. I think the results are interesting and definitely worth reporting, provided the authors can address the points below.

Major comments:

There is not enough information about the models. Minimally, the models should be described at a level of detail which would allow them to be reproduced. Ideally, software for the models themselves and the procedures for generating reward sequences would be available on the competition website.

The participants of the competition provided reward schedules (reward sequences), which are depicted in Fig. S1, and are fully described in the competition's code repository. Participants were not asked to describe in detail how these schedules were produced.

The CATIE and QL models are qualitatively explained in the text of the manuscript, described in detail in the supporting information section of the preceding paper Dan & Loewenstein (2019). In response to the reviewer's comment, we now added the software for running these models to the competition website:
<https://sites.google.com/view/cec19/Data>

Finding an optimal schedule given a model is no longer discussed in the revised manuscript. However, because it is discussed in the supporting information section of the preceding paper, we also added the optimization code to the website:
<https://sites.google.com/view/cec19/Data>

There is a lot of emphasis on CATIE (the winning model), but I'm not sure how justified this is given that the 2nd place model produced indistinguishable performance.

Thank you for highlighting this issue. In the revised manuscript, we now describe in more detail the principles that the architects of the second place used to design that

reward schedule, and the compassion of the first and second place. Specifically, in the Results section we write:

Second came a reward schedule that was submitted by a choice architect that combined primacy with strategic considerations to design the reward schedule (Table S1). Specifically, they created short streaks of trials in which only Alternative 1 is rewarded, followed by short streaks in which both alternatives were rewarded, in a hope that subjects will learn to exclusively choose Alternative 1, by that missing the rewards associated with Alternative 2. The average bias of that schedule was 64.1% $\pm$ 0.6% not significantly different from that of the first place ($p=0.84$, $df=1131$).

In the Discussion we write:

The design principles underlying the two most effective schedules were radically different. The winning schedule was engineered using a quantitative model of choice. The runner up, who was almost equally effective, was based on an intuitive understanding of human nature. The fact that the outcome of an automated optimization approach is as effective as the best intuition of academics is proof of principle of the potential of choice engineering.

Faced with an objective function and a quantitative model of human learning strategy, the winning choice engineer successfully designed an effective reward schedule using standard optimization techniques. This is an approach that can be automatically applied to other objective functions and constraints. By contrast, the choice architect approach that is based on human expertise and intuition cannot be easily automated and any change in the objectives and constraints would require the laborious design of a new schedule.

I'm surprised that the model comparison results in Fig. 4 differ so much between $E[p]$ and $E[\log(p)]$, given that $\log(p)$ is a monotonic function of p . Can the authors explain this?

Thank you for noticing and highlighting this interesting point. We address this point in our response to Reviewer 1 above (point 4, see also new Fig. S4)

The number of participants used to evaluate each model differs quite a bit. For example, Model 11 had 87 subjects, whereas Model 0 had 549 subjects. I'm sorry if I missed it, but I didn't see any justification of these differences. I suggest that the authors provide some evidence that these sample sizes are sufficient for reliable model comparison.

Thank you for commenting on this important issue. The substantial difference in sample size across different reward schedules is indeed intentional and calculated. The straightforward approach for estimating the performance of all the schedules in the competition would have been to allocate the same number of participants to all submissions. However, we had a slightly different goal. We were not so much interested in ranking the different submissions as we were interested in identifying with the greatest certainty the best schedule. With this goal in mind, if one finds that some of the submissions are very ineffective, future subjects should be allocated to better-performing schedules. The specific number of participants we used for each schedule was dictated by the static procedure “successive rejects”. This method “rejects”, in succession the least performing candidates, reallocating subjects to the remaining schedules. This procedure was already described in the announcement of the competition, and we elaborate on this issue in a supplementary titled “Supplementary text – Sample size”.

I was asked to comment on the dynamic track data, in particular whether it's reported in enough detail and whether it's worth incorporating into the manuscript. In my opinion, I think it makes sense to leave it where it is in the supplement. With respect to transparency, the reporting suffers the same issue as for the static track, namely the models are not described in sufficient detail to be reproducible.

Thank you for this comment. In terms of model description, we include in Table S2 the original language that the academic participants used to describe their models.

The competition's website now links to a code repository that contains the implementation of these models allowing their examination:

<https://drive.google.com/drive/u/2/folders/1uHacHfZJOFcmniuXhFEXM4FmY5mrE2B>

Minor comments:

Line 140: "subject's" -> "the subject's"

Thank you, we fixed this.

Fig 4: I assume the y-axis of panel b is probability x 100?

This was indeed a mistake, thank you for highlighting it.

Reviewer #3 (Remarks to the Author):

Thank you for sending me this paper for review. In it, Dan, Plonsky and Loewenstein describe the results of the Choice Engineering Competition (announced in Nat Comms in 2019) designed to show that quantitative models can be used to shape human behaviour. Briefly, the asked contestants to place a limited number of rewards such as to maximally bias behaviour. They obtained 11 submissions. They then tested each of these 11 submitted reward schedules on real human participants and examined them using two different types of prominent choice / learning models: a Q-learning model and a CATIE model. They distinguish choice engineering from choice architecture, with the former relying on an explicit model of choice; and the key findings are that contestants were able to design effective reward schedules; that CATIE outperformed Q learning in terms of explaining the human data (at least on some measures); and that CATIE also outperformed Q learning as a basis for designing effective reinforcement schedules.

There is much to like about the paper. It is an interesting and timely question which speaks to an important aspect of generalizability of a large swathe of in-lab studies. The project was clearly announced and the results should be of broad interest.

Overall, the paper is well written, but it would be good to be somewhat clearer about the nomenclature at several points, e.g. the distinction between architect and engineer; the distinction between the various datasets (lab, online, previous, new...); it would be helpful to have an overview figure over the process / datasets; and it would be good to have a brief table in the main paper describing the submissions as otherwise most of the figures are difficult to read.

Thank you for your comments.

Regarding the distinction between architect and engineer, this is an important point that was also raised by reviewer 1. In response, we rewrote the Introduction section, as is discussed in our response to Reviewer 1 (point 1).

Regarding the distinction between the various datasets, these different datasets are simply different behavioral records that were used to fit the QL model, resulting in different schedules. The relevancy of these datasets to the manuscript is secondary,

and we agree that they were confusing. Since these details were unnecessary for the understanding of the paper we omitted the names from the main text, and only discuss them in Table S1.

Regarding an overview figure, we tried various options but none seemed to improve the clarity of the manuscript.

Regarding a table describing the submissions, we feel that the details of most submissions do not contribute much to the understanding of the main text, and therefore prefer to leave them in the supplementary material, as Fig. S1 and with Table S1 explaining them.

Minor:

- Figure 3 seems to refer to 12 datasets (0-11), but there are only 11 elsewhere.

Thank you for raising this point. In the prior revision, the Results chapter opened with a presentation of the performance of Schedule 0, a schedule that was not part of the competition and was generated in an iterative process.

We fear that the presentation of the schedule in the previous version of the manuscript was confusing. In the revised version, following suggestions from the editorial team and other reviewers, the presentation of Schedule 0 is now delayed, so hopefully the distinction between this schedule and the ones used for the competition is clearer. While Schedule 0 was not included in the competition, its experiment was identical to the competition and thus we considered the data associated with it to answer questions that related to model comparison such as those presented in Figs. 3 and 4.

- some submissions refer explicitly to examining the results on prolific, which is also what the authors did themselves. it seems that this deserves some more attention - why did this not work?

Thank you for noticing and highlighting this issue. Schedule 11 does mention the use of Prolific participants. This could have been an interesting point. However, the authors of this schedule reached out and notified us, before the submission of

the manuscript, that unfortunately they identified an error in their implementation. We note this detail in Table S1, and attribute (only here) the poor performance of Schedule 11 to this technical problem.

- line 239: CATIE is stated to achieve 72.4 ± 0.1 , which is different from line 165 where it is 72.7 ± 0.1 .

Thank you for highlighting this issue.

The winning schedule (Schedule 1) achieved a bias of 72.4% on the CATIE model. We independently generated a schedule that was optimized on the same CATIE model. Because we used a slightly different optimization procedure, our “best” schedule for the CATIE model was slightly different from Schedule 1, and it achieved a comparable, but slightly different, bias.

Because this point is secondary to the main results of the manuscript, we removed it to enhance the clarity of the manuscript.

- figure 4: legend refers to light red, but there is no such colour in the figure. Please clarify error bars - quantiles would be more informative and allow relationship between b and c to be clearer.

We fixed the color indication.

Regarding the error bars, we now added that they correspond to the standard error of the mean.

Regarding quantiles, see our response to Reviewer 1 (point 4). Specifically, we added a new Fig. S4 to address this point.

- supplement table with models: hetro -> hetero,- supplement fig s2: scehulde -> schedule

Corrected. Thank you.